

Functional invariance of MLE

$$f_{\{x_i\}}(x; \sigma^2) = \prod_{i=1}^n \frac{e^{-(x_i - \mu)^2 / 2\sigma^2}}{\sigma \sqrt{2\pi}}$$

$$NLL = \sum_{i=1}^n \left\{ \frac{(x_i - \mu)^2}{2\sigma^2} + \log \sigma + \text{constants} \right\}$$

$$\frac{\partial NLL}{\partial \sigma} = 0 \rightarrow \hat{\sigma} = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2}$$

$$f_{\{x_i\}}(\{x\}; \nu) = \prod_{i=1}^n \frac{e^{-(x_i - \mu)^2 / 2\nu}}{\sqrt{2\pi} \sqrt{\nu}}$$

$$NLL = \sum_{i=1}^n \left\{ \frac{(x_i - \mu)^2}{2\nu} + \frac{1}{2} \log \nu + \text{constants} \right\}$$

$$\frac{\partial NLL}{\partial \nu} = \sum_{i=1}^n \left[\frac{(x_i - \mu)^2}{2} \frac{(-1)}{\nu^2} + \frac{1}{2\nu} \right] = 0$$

$$\sum_{i=1}^n \frac{(x_i - \mu)^2}{2\nu} = \frac{n}{2\nu}$$

$$\hat{\nu} = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2$$

$$W = g(\sigma)$$

$$\hat{W} = g(\hat{\sigma})$$

$$\hat{W} = \text{MLE of } W$$

$$\hat{\sigma} = \text{MLE of } \sigma$$

$$\frac{1}{\sigma} \frac{e^{-(x - \mu)^2 / 2}}{\sqrt{2\pi}}$$

$$\rho = 1/\sigma^2$$

inverse variance
or "precision"

$$X \sim \text{Bernoulli}(\theta)$$

$$f_X(x) = \theta^x (1-\theta)^{1-x} \quad x \in \{0,1\}$$